

Shared evidence | Core model cards

Blood glucose: build the control model

Name _____ Date _____

A qualitative classroom model. There are no medical concentration values, timings or treatment rules.

A

Glucose is absorbed into blood; concentration rises.

B

Pancreatic beta cells release more insulin.

C

Insulin-responsive tissues increase glucose uptake/storage processes.

D

The concentration tends towards the usual suitable range; the insulin stimulus reduces.

Role reminders: insulin is a hormone signal. Glucose is a sugar used by cells. Glycogen is a carbohydrate store made from joined glucose units. Some glucose uptake is insulin-independent; the model focuses on insulin-responsive tissues.

W1. Predict and order the core response. W2. Correct the substance swap. W3. State a model limit.

Supported route | Trace the corrective response

Word bank: pancreas; insulin; blood; glucose; glycogen; uptake; towards.

S1. Order cards A-D and explain why B follows A.

S2. Identify the signal, the sugar and the store. Use insulin, glucose and glycogen once each.

S3. Explain how the target response can reduce raised blood glucose. Name one target tissue.

S4. Why does the extra insulin stimulus reduce as glucose returns towards usual levels? Give one limit of the model.

Standard route | Explain the control loop

Use mechanisms rather than a story about a hormone choosing what to do.

M1. Explain the sequence from glucose absorption to pancreatic insulin release and target-tissue response.

M2. Distinguish insulin, glucose and glycogen, including why insulin does not become the stored material.

M3. Explain why this is a corrective feedback response, not an endlessly increasing output.

M4. Evaluate a claim that the model predicts an exact blood-glucose concentration over time. Identify two missing kinds of information.

Stretch route | Test the limits and optional second branch

Optional deeper evaluation. T3 is Higher-tier only: lower blood glucose can stimulate pancreatic alpha cells to release glucagon; the liver releases glucose, including by glycogen breakdown.

T1. Explain why “all cells need insulin for all glucose uptake” is too broad. Give a better core statement.

T2. The same qualitative rise is shown for two fictional people. Must their timing or exact concentrations match? Explain.

T3. Optional Higher: sequence lower glucose → glucagon release → liver response → towards usual. Contrast it with the insulin branch.

T4. Suggest one feature this model would need to compare real measurements. Explain why adding that feature alone would not validate its predictions.

Exit and reflection | A signal is not a sugar

Use precise words or labelled arrows.

E1. Which organ releases insulin into blood?

E2. Which material supplies the units stored as glycogen?

E3. What limits this model to a qualitative explanation rather than an exact concentration prediction?

L1. What did your audience question? Improve a label or causal explanation.

Next useful step

- A. Sequence the insulin response.
- B. Separate hormone, sugar and store.
- C. Explain corrective feedback.
- D. State a model limitation.